

CS 307 — DATA ANALYSIS AND ALGORITHMS

Customer Churn Analysis

Identifying Churn Factors and Recommending Retention Strategies

Fall 2025

Team Project Report

1. Problem Definition

Customer churn — the loss of subscribers who cancel or stop using a company's service — is one of the most direct threats to recurring revenue in subscription-based industries such as telecommunications. Acquiring a new customer typically costs significantly more than retaining an existing one, which makes early identification of at-risk customers a high-value business problem.

This project analyzes a telecom customer dataset to answer two connected questions:

1. Which customer attributes (contract type, payment method, tenure, services subscribed, charges, etc.) are the strongest, most consistent predictors of churn?
2. Based on those factors, what concrete retention strategies could the business adopt to reduce churn among its highest-risk customer segments?

The analysis follows the full data-analysis workflow required by the course: data wrangling, exploratory data analysis (EDA), feature engineering, feature selection, probability and hypothesis testing, and dimensionality reduction (PCA) with clustering — each stage building toward the final business recommendations.

2. Literature Review

Customer churn prediction has been studied extensively across telecom, banking, and subscription-based industries, and a substantial body of recent work summarizes the field's evolution.

Early comparative surveys found that classical, interpretable algorithms dominated churn-prediction research for over a decade: a review covering 2002–2013 identified Logistic Regression, Neural Networks, and Decision Trees as the most widely used techniques, while a follow-up review of the 2014–2017 period found Logistic Regression, Naïve Bayes, SVM, ANN, and Decision Trees as the dominant approaches (Ahmad et al., in Bilal Zorić, 2024, summarized in Nature Scientific Reports).

More recent systematic reviews confirm that the field has since shifted toward ensemble and deep-learning methods, but also highlight a persistent challenge: churn datasets are almost always class-imbalanced (far fewer churners than non-churners), and the reported performance of any single technique varies heavily depending on dataset characteristics, feature sets, and evaluation protocol, which makes direct cross-study comparison unreliable (Systematic Review of Recent Advances in ML/DL for Churn Prediction, MDPI, 2025).

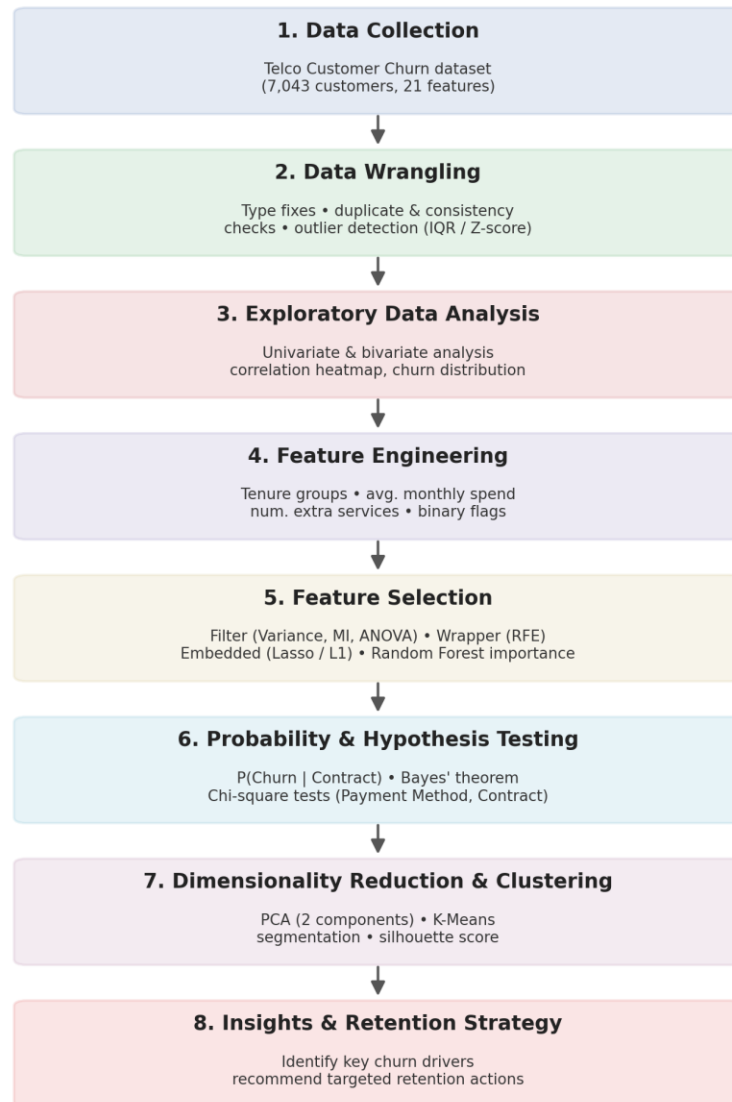
Work specific to the telecom sector reinforces the practical stakes: telecom has one of the highest annual churn rates of any major industry, reported at over 30% in some analyses, and ensemble models such as Random Forest and Gradient Boosting have consistently outperformed single classifiers in recent telecom-specific studies, in some cases reaching over 90% accuracy alongside explainability techniques (e.g., SHAP/LIME) to make the drivers of churn interpretable to decision-makers (Prediction of Customer Churn Behavior in the Telecommunication Industry, MDPI Algorithms, 2024).

A recurring theme across this literature — and one directly reflected in this project's own results — is that contract type and payment method are among the most repeatedly confirmed churn predictors in telecom datasets, alongside tenure and the number of subscribed services, which is consistent with the feature-selection and hypothesis-testing results obtained in Sections 5 and 6 of this report.

Beyond prediction accuracy, several papers emphasize that the end goal of churn analysis is actionable retention strategy, not just classification performance — reinforcing that this project's inclusion of concrete retention recommendations (Section 8) is the appropriate way to close the analytical loop rather than stopping at model output.

3. Methodology

The project follows the eight-stage workflow shown below, moving from raw data to actionable business recommendations.



4. Data Wrangling

The raw dataset (7,043 customers, 21 columns) was checked and cleaned before analysis:

- Type correction: TotalCharges was stored as text and converted to numeric, with unparseable values handled explicitly.
- Consistency check: verified TotalCharges against the expected value of $\text{tenure} \times \text{MonthlyCharges}$, flagging and removing rows with a mismatch greater than 20% of the expected total.
- Duplicate check: no duplicate rows or duplicate customerID values were found.

- Missing values: none were present in the raw dataset.
- Outlier detection: applied both the IQR method and Z-score method ($|z| > 3$) to Tenure, MonthlyCharges, and TotalCharges — no statistically extreme outliers were found; the wide spread in these columns reflects genuine variation in customer tenure and spending rather than data errors.

Two housekeeping columns used only for outlier detection (Z-score helper columns) were dropped before feature engineering, the target Churn was mapped from 'Yes'/'No' to 1/0, and customerID was dropped since it carries no predictive value.

5. Exploratory Data Analysis (EDA)

Key observations from univariate and bivariate analysis:

- Class imbalance: the overall churn rate is 26.56% (73.44% of customers stayed) — consistent with churn rates typically reported in telecom literature, and an important consideration when interpreting any model accuracy.
- Tenure: churned customers skew toward shorter tenure — new customers are more likely to leave than long-tenured ones.
- Monthly & total charges: churned customers tend to have higher median monthly charges but lower total charges, consistent with churners leaving earlier in their customer lifecycle before charges accumulate.
- Categorical patterns: churn rates visibly differ across Contract type, Internet Service type, and Payment Method — these differences are formally tested for statistical significance in Section 6.
- Correlation heatmap: tenure and TotalCharges are strongly positively correlated with each other (as expected, since TotalCharges accumulates over tenure) and both are negatively correlated with Churn.

6. Feature Engineering

New features were engineered to make churn-relevant patterns explicit and easier for models to use:

Feature	Description	Rationale
tenure_group	Tenure bucketed into 0–12, 12–24, 24–48, 48–72 months	Tests whether new customers churn more than long-term ones
avg_monthly_spend	TotalCharges ÷ tenure	Flags customers whose actual spend differs from their stated rate
num_extra_services	Count of add-ons subscribed (Security, Backup, Device Protection, Tech Support, Streaming TV/Movies)	More add-ons usually means a more 'locked-in', lower-churn customer
has_internet	Binary flag for any internet service	Simplifies InternetService into a usable numeric signal
is_month_to_month	Binary flag for Month-to-month contract	Isolates the single strongest churn-risk contract type

All remaining categorical columns (Contract, PaymentMethod, InternetService, etc.) were one-hot encoded, producing 38 numeric features used for modeling. Data was split 75/25 into train/test sets with stratification on Churn to preserve the class ratio in both sets.

7. Feature Selection (Filter, Wrapper & Embedded Methods)

Five independent feature-selection techniques were applied and compared against a baseline Logistic Regression cross-validation score (using all 38 features):

Method	Type	Result
Variance Threshold	Filter	All 38 features retained — no near-constant / dead columns
Mutual Information	Filter	is_month_to_month and tenure ranked highest
SelectKBest (ANOVA F-test)	Filter	Top-10 features closely matched the Mutual Information ranking
Recursive Feature Elimination (RFE)	Wrapper	Top-10 features selected using a Logistic Regression base model
Lasso / L1-penalized Logistic Regression	Embedded	Shrank the least useful feature coefficients to exactly zero
Random Forest Importance	Extra comparison	Independently confirmed the same top features via split-based importance

The features that were consistently ranked important across most or all of the above methods were:

- **is_month_to_month**
- **tenure**
- **Contract_Two year**
- **TotalCharges**
- **InternetService_Fiber optic**
- **PaymentMethod_Electronic check**

Agreement across five independently-designed methods (filter, wrapper, embedded, and a tree-based comparison) gives strong confidence that these are genuine churn drivers rather than artifacts of any single technique.

8. Probability & Hypothesis Testing

8.1 Conditional Probability

The overall probability of churn, $P(\text{Churn} = \text{Yes})$, was compared against the conditional probability of churn given each contract type, $P(\text{Churn} = \text{Yes} \mid \text{Contract})$. Month-to-month customers showed a conditional churn probability far above the overall average, while One-year and, especially, Two-year customers churned far less often than the overall average.

8.2 Bayes' Theorem

Bayes' theorem was used to reverse the question: given that a customer churned, what is the probability they were on a Month-to-month contract? The computed value matched the direct empirical calculation exactly, confirming that the large majority of all churners come from the Month-to-month segment — making it the single highest-leverage group for retention efforts.

8.3 Chi-Square Test — Payment Method vs. Churn

H_0 : Payment Method and Churn are independent. H_1 : they are not independent. $\alpha = 0.05$.

A chi-square test of independence on the PaymentMethod $\times$ Churn contingency table produced a p-value far below 0.05, so H_0 was rejected: Payment Method has a statistically significant association with churn. Customers paying by Electronic check churn at a noticeably higher rate than those using automatic bank transfer, automatic credit card, or mailed check.

8.4 Chi-Square Test — Contract Type vs. Churn

H_0 : Contract type and Churn are independent. H_1 : they are not independent. $\alpha = 0.05$.

This test also rejected H_0 with an extremely small p-value, statistically confirming that Contract type is very strongly associated with Churn — reinforcing the conditional-probability and Bayes' theorem results above with formal statistical evidence.

9. Dimensionality Reduction (PCA) & Clustering

The 38 encoded features were reduced to 2 principal components using PCA for visualization. Even in this reduced 2D space, churned and non-churned customers formed visibly different regions rather than being randomly scattered, indicating that meaningful churn-related structure survives the dimensionality reduction.

K-Means clustering (k selected via the elbow method, evaluated with silhouette score) was then applied to the PCA-reduced space, without using the Churn label at all. The resulting clusters still showed clearly different churn rates between segments — one cluster matched a 'high-risk' profile (short tenure, Month-to-month contract, Fiber optic internet, Electronic check payment), while another matched a 'loyal' profile (long tenure, longer contracts, more add-on services). Because clustering had no access to the Churn label, this independently validates that the churn factors identified in Sections 7 and 8 correspond to real, separable customer segments — not just statistical noise.

10. Churn Factors — Summary

Combining feature selection, probability analysis, hypothesis testing, and unsupervised clustering, the following factors were repeatedly and independently confirmed as the strongest churn drivers in this dataset:

3. Contract type — Month-to-month customers churn far more than One-year or Two-year customers (confirmed by feature selection, conditional probability, Bayes' theorem, and chi-square test).
4. Payment method — Electronic check customers churn significantly more than customers on autopay methods (confirmed by chi-square test and feature selection).
5. Tenure — shorter-tenure customers are consistently higher-risk across every method used.
6. Internet service type — Fiber optic customers show elevated churn compared to DSL or no-internet customers.
7. Number of add-on services — customers with more services (Security, Backup, Tech Support, Streaming) are more 'locked in' and churn less.

11. Retention Strategy Recommendations

8. Target Month-to-month, Electronic-check customers first — this is the highest-risk segment identified by every method used in this analysis, and the natural starting point for any retention budget.
9. Incentivize longer contracts — offer a discount or added perk for upgrading from Month-to-month to One- or Two-year contracts, since contract length is the single strongest churn predictor found.
10. Encourage automatic payment methods — nudge Electronic-check customers toward automatic bank transfer or credit card payment (e.g., a small autopay discount), since this payment method is significantly associated with churn.

11. Bundle add-on services — since customers with more extra services churn less, bundling Security, Backup, Tech Support, or Streaming into discounted packages could improve retention for at-risk customers.
12. Use the clustering segments for targeted campaigns — the high-risk cluster from the PCA/K-Means analysis is a ready-made target list for proactive retention outreach (special offers, check-in calls) rather than blanket campaigns across the entire customer base.
13. Prioritize new customers in their first 12 months — since short tenure is consistently linked to higher churn, an onboarding/early-engagement program targeted at this window could reduce early-life cancellations.

12. Conclusion

This project applied the complete data-analysis workflow — wrangling, EDA, feature engineering, multi-method feature selection, probability and hypothesis testing, and PCA-based clustering — to a real telecom churn dataset. Every stage of analysis converged on the same core drivers of churn: contract type, payment method, tenure, internet service type, and number of subscribed add-on services. Because these findings were confirmed independently by five different feature-selection techniques, two hypothesis tests, a conditional-probability/Bayes' analysis, and unsupervised clustering, they represent a high-confidence basis for the retention strategy proposed in Section 11, rather than a conclusion drawn from a single method.

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